A-A-0051580 (DM)
4 August 1986

INTERIM COMMERCIAL ITEM DESCRIPTION

INVESTMENT, PHOSPHATE-BONDED MATERIAL

The General Services Administration has authorized the use of this Interim Commercial Item Description which was developed by DLA, Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, based upon currently available technical information.

This interim commercial item description covers a phosphate-bonded investment in the casting of precious and non-precious dental alloys with either the thermal or hygroscopic expansion techniques. The investment material shall be Whip-Mix's Ceramigold phosphate-bonded investment material, or equal, with the following salient characteristics:

Salient characteristics:

The phosphate-bonded investment material shall be used to cast precious and non-precious dental alloys intended for porcelain application in both thermal and hygroscopic techniques. The investment shall consist of a powder and a silicate liquid. The powder material shall be essentially a composition of phosphate-bonded material with carbon and other chemical modifiers. Thermal and setting expansion shall be variable based, on water dilution of the liquid, to give a combined thermal and setting expansion of over 2% to below 1%. The investment material shall contain no gypsum and shall not liberate gaseous sulfur. The color of the investment material shall be of a grey color. As an option, a 1-1/8 inch wide flask liner shall be provided. The physical properties required for the investment material shall be as specified in table I.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Physical Properties

Mixing Consistency	16ml liq to 100g powder
Working time	2 - 4 minutes
Setting time	4 - 6 minutes
Compressive strength(min)	63kg/cm^2
Hygroscopic expansion	1-1/2% - 2%
Ready for burnout	45 minutes
Setting expansion @ 25 ± 2°C	
Minimum percent	0.80
Maximum percent	1.20
Thermal expansion	
Temperature degrees (Centrigrade)	700°C to 1000°C
Maximum percent	1.44
Minimum percent	0.96

Adequate and accurate instructions for proportioning and manipulating dental casting investment shall be furnished with each unit package. The package shall consist of 144 sixty gram envelopes of powder, with 1,240 milliliters of silicate liquid. The 60 grams of investment shall be packaged in a waterproof, laminated foil material, envelope of appropriate size.

Workmanship. The phosphate-bonded investment shall be dry, free from set or caked particles, uniform in texture and free of impurities which may impair the satisfactory performance of the investment. Coloring matter, as such, shall not be regarded as an impurity.

Unit. Each (EA). Each unit consisting of 144 sealed 60 gram envelopes along with sufficient liquid, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Inspection shall be conducted to determine compliance with specification requirements.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one unit. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractor/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

Stock Number	Item Identification
6520-01-162-2010	INVESTMENT, DENTAL, Phosphate Bonded, 144 Sixty-Gram Envelopes, with Diluent.

Ordering data. (Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD - MB

Agent:

DLA - DM

CIVIL AGENCY COORDINATING ACTIVITY
VA-OSS

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